Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/26/2025
PI Name:	Wood, Scott J. Ph.D.		
Project Title:	Assessments for Lunar Landing and Egress During Early Artemis Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations (3) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:	NOTE: PI returned to NASA JSC in January 2017. PI was at Azusa Pacific University from August 2013 – January 2017; prior to August 2013, PI was at NASA JSC.		
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Start date changed to 01/10/2025 per C. Ribeiro/HHC (Ed., 12/17/2025).		
Key Personnel Changes/Previous PI:	N/A		
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Grant/Contract No.:	Directed Research
Performance Goal No.:	
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Task Description:	This purpose of this directed study is to strategically capture data during the early Artemis test flights to better characterize individual and mission design factors that may impact human performance during critical mission tasks. This study will specifically target the initial missions that will return humans to the lunar surface, Artemis III -- V currently scheduled in the 2027 -- 2029 timeframe. Our objectives are organized around key mission phases of preparing for landing, lunar descent and ascent, and early surface operations. While subject participation will be limited, this initial Artemis study will address critical human system risks (sensorimotor, extravehicular activity and cardiovascular) identified for lunar surface missions. As we build on the prior evidence from the Apollo missions, it is important to note there are some key differences with the planned Artemis missions that may significantly impact performance on tasks during or after these g-transitions. These include longer transits in microgravity which may increase deconditioning effects, a more diverse astronaut cohort, differences in landing vehicle design that impact crew override capabilities, increased distance from the rendezvous vehicle, and more challenging terrain lighting conditions associated with the planned South Pole landing sites. Therefore, it is critical to capture data on these early Artemis missions to help enhance our ability to plan, train and fly future exploration programs. Aim 1: Preflight and On-Board Training Assessments for Lunar Landings Preflight training across a continuum of simulation platforms during Apollo was critical for mastery level proficiency and handling of disorientation challenges during landing, e.g., visual illusions from blowing dust (Woodling et al., 1973). Our first aim is targeted to enhance the Artemis Human Landing System (HLS) landing proficiency through preflight motion simulations and on-board training. Aim 1a will be to refine existing spatial orientation models utilizing an observer approach to predict perception during HLS landing profiles with the goal of improving preflight piloting training in motion-based simulation platforms. Mathematical models for spatial orientation perception have been used to improve the design of control algorithms used for motion cueing with this motion base (Dixon et al., 2019). Using crew perceptual feedback during the HLS landings, we intend to refine the models to improve the motion simulation fidelity (Love et al., 2025). Aim 1b will be to utilize an on-board tablet-based trainer in an HLS-relevant supervisory landing task. As mission duration increases, lack of training retention is expected to negatively impact operator performance of complex tasks (Foale et al., 2005; Dempsey and Barshi, 2021). The tablet task will be performed by lunar surface subjects and NRHO (Near Rectilinear Halo Orbit) control subjects three times each during transit to the Moon prior to docking with the HLS vehicle. A fourth session will be added if the transit is an extended duration and/or scheduling allows. The tablet task is patterned after the Multi-Attribute Task Battery (Cegarra et al., 2020) and is currently being used as part of the Manual Piloting study on the International Space Station (ISS). Aim 2: Characterizing Responses during Lunar Descent and Ascent Our second study aim is to characterize crew perceptual responses and physiological parameters during lunar descent and ascent to characterize sensorimotor, cardiovascular, and other individual and landing trajectory factors that may impact landing, early surface egress and docking performance. While this type of data recording was limited during prior Apollo missions (Petersen et al., 2020), this aim is critical to explore differences in responses across vehicle providers (SpaceX Starship during Artemis III and IV versus Blue Origin's Blue Moon during Artemis V) which involve different vehicle trajectories, different position of the crew relative to the vehicle center of mass, and importantly different postures (seated versus standing). Crew activity during the descent and ascent phases will be characterized by a combination of markerless motion capture using in-cabin video and body-worn inertial measurement unit (IMU) acceleration as available, and user-interface interactions with displays and/or inceptors for controlling vehicle state. Head motion during descent and ascent coupled with vehicle accelerations and crew perceptions during post-lunar landing debriefs will refine spatial orientation modeling predictions (2a). The second part of this aim (2b) is to record continuous recording of heart rate or EKG from their intravehicular activity (IVA) suit synchronized with seat acceleration data, suit thermal load data, and mission events to validate acceleration limits and countermeasures after adaptation to microgravity (descent) and partial-G (ascent). Cardiovascular responses to accelerations during these dynamic flight phase will aid in defining future limits, with and without orthostatic countermeasures. Crew medical data will be leveraged to capture inflight exercise and fitness assessments as well as operational interventions (e.g., medications, fluid loading) conducted around these critical mission phases. Aim 3: Post-Landing Lunar Surface Assessment Battery Our third study aim is to characterize sensorimotor, cardiovascular, and metabolic responses in crewmembers on the lunar surface using a post-landing lunar surface assessment battery. The aim contains two parts: a) a sensorimotor EVA assessment test battery, and b) a cardiovascular test battery. Our aim 3a test battery will be based on a set of sensorimotor assessments currently being validated with ground-based analogs (Moudy et al., 2023; Moudy et al., 2024). The test battery will include tasks involving motion sensitivity, obstacle avoidance, kneel-and-turning, and fall recovery. Motion sensitivity, assessed via subjective perception of movement during head motion, will further refine the Aim 1a spatial orientation perception model. In addition to performance metrics such as time to complete or kneeling limits of stability and subjective ratings, metabolic responses (e.g. heart rate) during EVAs and markerless motion capture and IMU data will be used as available. Aim 3a would be conducted twice preflight, and a maximum of four times on the lunar surface: as soon as possible after landing, once or twice mid-way through the surface operation timeline, and prior to ascent. For aim 3b, a cardiovascular test will be adapted from previous ground-based implementations (Lee et al., 2020) to characterize the cardiovascular response to an orthostatic stress, quantify the fluid shift in varying postures during 1/6 G , and assess the status of venous flow that may increase risk for venous thrombus to inform future con ops (medical kit) and countermeasures. The test will require an operator and continuous heart rate measurement, 2-D and Doppler ultrasound capability with data/video recording, and brachial blood pressure device from Med Kit with recording (e.g., data log or audio). This procedure involves measuring internal jugular vein cross-sectional area and Doppler flow along with heart rate and blood pressure, first while supine, then during standing which follows prone rest, and finally seated. The assessments performed in transit/orbit will be identical but without the changes in posture. Aim 3b would be conducted twice preflight, once during transit to the Moon, once or twice during lunar surface operations, once during transit back to Earth, and once postflight. Data sharing will be utilized with Standard Measures and medical tests. These data will be compared to metrics obtained during these assessments with EVA crew debriefs and metabolic data obtained during the early lunar EVAs. Crewmembers that will not travel to the lunar surface (i.e., will remain in orbit around the moon) will be recruited as control subjects to perform the cardiovascular measures (Aim 3b) in a microgravity environment only.

Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This research is directed because there is insufficient time for solicitation.
Research Impact/Earth Benefits:	This study is expected to provide key deliverables for the Moon-to-Mars program for both Artemis Lunar and future Mars missions. The perceptual models developed for Aim 1a will provide a basis for future preflight simulation training as well as a means of evaluating the spatial disorientation risks for future vehicle trajectories. The aim 1b inflight on-board training assessments will provide a countermeasure for manual crew override performance during longer microgravity transits. Aim 2 will help characterize descent and ascent risks related to sensorimotor and cardiovascular deconditioning associated duration of microgravity transit to lunar surface, including validation of existing countermeasures. The Aim 3a will validate a post-landing assessment battery used to inform early EVA preparedness. Aim 3b will help characterize headward fluid shift and venous flow status during exposure to lunar gravity, and provide information about whether 1/6 G will provide any protection from spaceflight risks associated with the headward fluid shift.
Task Progress:	Following the successful peer and administrative review, the Notice of Intent to Award was received with funding which started in January 2025. The Feasibility Assessment for Artemis 3 was initiated with the Research and Operations (ROI) element using the initial proposal in January 2025. At the request of HHC, a revised proposal was submitted in late March that added control subjects from the Near Rectilinear Halo Orbit (NHRO) crew members for some measures, as well as placeholders for additional measures during lunar surface operations as opportunities arise. The initial IRB was approved in May 2025. The following highlights some of the accomplishments and ongoing activities during this initial reporting period for the specific aims. Aim 1: Preflight and On-Board Training Assessments for Lunar Landings Related to our perceptual modeling goals (Aim 1a), significant progress has been made this past year under the NASA HPR-Navy Interagency Agreement (IAA) by Dr. Clark, University of Colorado Boulder, during the Phase 1 testing of the lunar landing simulations at the Disorientation Research Device (DRD, aka Kraken) in support of the development of the lunar landing skills training for the Flight Operations Directorate. Dr. Clark integrated the vehicle state data provided by Langley from the Government Reference Vehicle with the Observer Model as a means of validating the motion cueing from the motion simulation data provided by the DRD team. While Dr. Clark did not specifically capture perceptual responses from the test pilot participants during the Phase 1 testing, we plan to extend Dr. Clark's work using subjective questionnaires and computer animations during ascent and descent simulation and flight activities in an effort to further validate the computational modeling for the lunar gravity environment. The initial set of descent and ascent debrief questions was formulated, and we have begun to explore the feasibility of using animations in a similar manner as Holly et al. (2019). Near-term work will include defining the vehicle trajectory data that will be requested from the HLS vendors via the Payload Integration Agreements and developing the crew training plan to train the flight crew on the perceptual reporting requirements. This will be done using the same landing simulation as integrated in the DRD that has been installed in the Neuroscience Lab hexapod. Additional training opportunities will be explored during the DRD skills-based training in Dayton under the NASA-Navy IAA. We anticipate that the enhanced modeling efforts from this specific aim will be used to improve future motion-based preflight landing simulations and training, as well as to evaluate the spatial disorientation risks associated with future vehicle landing trajectories. Significant experience this past year has been gained on the Aim 1b Tablet Task, named ATLAS (Weiss et al., 2025), as part of the Manual Piloting study. Lessons learned from both ISS and ground studies will help characterize the learning effects of this supervisory landing task as well as refine the trial metric scorecard used inflight. Collaborators at the Johnson Space Center (JSC) ER Simulation and Graphics Branch will be porting the ISS iOS application to Windows for deployment on Orion Surface Pros. The application will be enhanced to allow for custom landing playbacks that are specific to each Artemis mission, with expanded post-trial metrics to promote on-board training during the Orion transit. Near-term work will include reviewing this application with the HLS Crew Training Officers (CTOs) to ensure the application will build on more vendor-specific onboard trainers as the crew transitions to the HLS vehicle and not pose a risk of negative training. This specific aim was expanded to include the orbiting NRHO (Near Rectilinear Halo Orbit) crew as control subjects as part of assessing changes in the cognitive workload measures during adaptation to weightlessness during the Earth-to-lunar transit. Aim 2: Characterizing Responses during Lunar Descent and Ascent One of the more challenging tasks of this study is to capture crew activity during the descent and ascent phases. We have been working with the Research and Operations Integration (ROI) element to evaluate the feasibility of body-worn inertial measurement units (IMUs) or tri-axis accelerometers. We have proposed IMUs developed by Diversified Technical Systems (DTS, Seal Beach, CA) that were part of a prior NASA Small Business Innovative Research (SBIR) grant guided by the Applied Injury Biomechanics laboratory at NASA. It is assumed that the IMUs currently used in the SpaceX Dragon vehicle will be similar to those integrated in the Starship seats for impact load monitoring, although those details are not yet confirmed. Since it is unclear if motion sensors can be certified for in-suit monitoring, markerless motion capture using in-cabin video is also under review. We anticipate identifying specific cabin views and other sampling characteristics (e.g., start/stop of recording, sample rate) that may enable offline processing to characterize crew activity during the descent and ascent phases. This data will then be integrated with the vehicle trajectory data as input into the perceptual models from Aim 1a as a basis for refining the models based on crew feedback. We are evaluating how data shared across HLS Technical Objectives, e.g., impact loads analysis, can be used to further refine the PIA for joint measures of head and body kinematics. We anticipate the cardiovascular monitoring (Aim 2b) will also rely heavily on both medical and operational data sharing to capture continuous recording of heart rate, suit thermal load data, medical interventions, and mission events to validate acceleration limits and countermeasures after adaptation to microgravity (descent) and partial-G (ascent). Aim 3: Post-Landing Lunar Surface Assessment Battery Significant work this past year involved the refinement of the Artemis Mission Medical Evaluation Requirements Document (AMMERD) proposed measurements guided by the Artemis Research and Operations Working Group under the direction of both the JSC Human Health and Performance Directorate and the Human Research Program. Our team has worked to ensure alignment of the AMMERD 7.1 Pre/Post EVA assessments with our Aim 3a Sensorimotor Assessments. These measures have been approved for consideration by the Aerospace Medicine Board and the Exploration Integration Panel, prior to final disposition by the multilateral medical working groups. We are leveraging lessons learned from ground studies using various analogs and are exploring potential lunar parabolic opportunities. While we anticipate these assessments will utilize simple subjective scales and timed measures for real-time medical decision making, we plan to utilize the markerless motion capture and

	IMUs described in Specific Aim 2 for post-mission processing. For the Aim 3b Cardio Assessments, the ROI engineering team down-selected to two ultrasounds (Phillips Lumify and GE Vscan Air) and two tablets (Samsung Active 10” and Galaxy 9A 11”, both Android) based on testing at the NASA space radiation laboratory (NSRL, Huff et al., 2023). During the coming year, we will be developing procedures for the various mission phases using these two systems. This work will continue to be coordinated with AMMERD 1.9 ultrasound imaging measures. This specific aim was expanded to include the orbiting NRHO crew as control subjects.
Bibliography Type:	Description: (Last Updated: 02/20/2026)